

Aryan Thodupunuri

US Citizen | 571-442-7810 | aryan20544@gmail.com |  Portfolio |  LinkedIn |  GitHub

EDUCATION

University of Virginia

B.A. in Computer Science, Accelerated M.S. in Data Science | **GPA: 3.90/4.00**

Charlottesville, VA

Expected May. 2027

EXPERIENCE

Amazon Web Services (AWS)

Arlington, VA

Solutions Architect Intern (AI/ML)

Summer 2026

- Building an **AI-assisted developer platform** that helps engineers navigate large codebases by combining **AWS Bedrock, vector search, and retrieval workflows** to preserve context across complex software projects.
- Designing **retrieval and evaluation pipelines** for semantic search, context compression, and code understanding to improve the quality and reliability of developer-facing AI systems.
- Developing production-grade systems in Python and AWS leveraging **5+ cloud services and retrieval-based architectures**, while collaborating with engineers to improve reliability, performance, and developer productivity.

Booz Allen Hamilton

McLean, VA

Software Engineering Intern

May 2025 – Aug. 2025

- Developed a **cloud-native supplier intelligence platform** using **React, Flask, AWS, and PostgreSQL** to centralize **10K+ vendor records** and streamline research workflows for **50+ analysts**.
- Engineered a production **RAG pipeline** using **vector embeddings, semantic retrieval, reranking, and evaluation** to reduce document review to ~5 seconds.
- Developed **Python and C++ data processing pipelines** to extract, validate, and structure information from PDF contracts, improving data quality and supporting large-scale analytics workflows.

Red Light Management

Charlottesville, VA

Software Engineering Intern

Sept. 2024 – May 2025

- Built **high-throughput** ingestion pipelines processing **50K+ daily events** from Spotify and Ticketmaster **REST APIs** to drive actionable downstream business insights and real-time analytics for artist management teams.
- Improved backend performance by **35%** with **Redis caching**, load balancing, and low-latency infrastructure

UVA Link Lab

Charlottesville, VA

Machine Learning Engineering Intern

April 2024 – Aug. 2024

- Developed machine learning models in **TensorFlow** using **CNN-RNN architectures** to analyze spatial and time-series flood patterns, improving forecasting accuracy by **15%** over traditional approaches.
- Evaluated model performance using **statistical validation, RMSE, MSE, and experimentation workflows**, reducing runtime by **95%** compared to simulation-based baselines.

PROJECTS

CloudSentinel | *Python, FastAPI, AWS, PostgreSQL, ChromaDB*

April 2026 - May 2026

- Built AI-powered cloud analytics platform using **FastAPI, AWS, PostgreSQL, and agentic workflows** to analyze cloud spending, detect anomalies, and generate infrastructure optimization recommendations.
- Engineered **retrieval and indexing pipelines** using **vector embeddings, ChromaDB, CloudWatch, and AWS Cost Explorer APIs** to transform operational cloud data into searchable insights.

CourseCompass Chrome Extension | *Java, Spring Boot, PostgreSQL, GraphQL*

March 2026

- Engineered an extension serving **25,000+ student queries** by aggregating RateMyProfessors and Reddit data via parallel data fetching with **Java CompletableFuture** to achieve sub-500ms aggregation and caching latency.
- Built a scalable backend using **PostgreSQL, Redis, and GraphQL** that reduced database load by **40%** during peak registration periods while maintaining low-latency query performance.

TECHNICAL SKILLS

Languages: Python, Java, SQL, Go, C++, JavaScript, TypeScript, Shell, GraphQL

Backend & Software Engineering: Object-Oriented Design, Data Structures & Algorithms, REST APIs, Microservices, Concurrent Programming, Distributed Systems, SDLC

Database & Data Systems: PostgreSQL, Redis, SQL, GraphQL, ChromaDB, Vector Databases

Cloud & Infrastructure: AWS Bedrock, EC2, Lambda, S3, RDS, IAM, Docker, Terraform, Kafka, CI/CD

AI & Retrieval Systems: RAG, Vector Embeddings, Semantic Search, Retrieval Systems, Model Evaluation, LLMs

Certifications: AWS Certified Developer - Associate, AWS Solutions Architect – Associate, AWS AI Practitioner